

EX NO:2	Chebyshev IIR Filter
DATE:	2.1. Design and Performance Evaluation of a Chebyshev Type-I IIR Low-Pass Filter using MATLAB

CODE :-

```

clc;
clear;
close all;
% Specifications
Wp = 0.4;
Ws = 0.55;
Rp = 1;
Rs = 40;
% Determine filter order
[N,Wn] = cheb1ord(Wp,Ws,Rp,Rs);
% Design Low-Pass Filter
[b,a] = cheby1(N,Rp,Wn);
% Display Results
fprintf('Filter Order = %d\n',N);
fprintf('Cutoff Frequency = %.2f * pi rad/sample\n',Wn);
if all(abs(roots(a))<1)
    disp('System is STABLE');
else
    disp('System is UNSTABLE');
end
%% Magnitude & Phase Response
figure;
freqz(b,a,1024);

```

```
title('Magnitude and Phase Response');
```

```
%% Impulse Response
```

```
figure;
```

```
impz(b,a);
```

```
grid on;
```

```
title('Impulse Response');
```

```
%% Group Delay
```

```
figure;
```

```
grpdelay(b,a);
```

```
grid on;
```

```
title('Group Delay');
```

```
%% Pole-Zero Plot
```

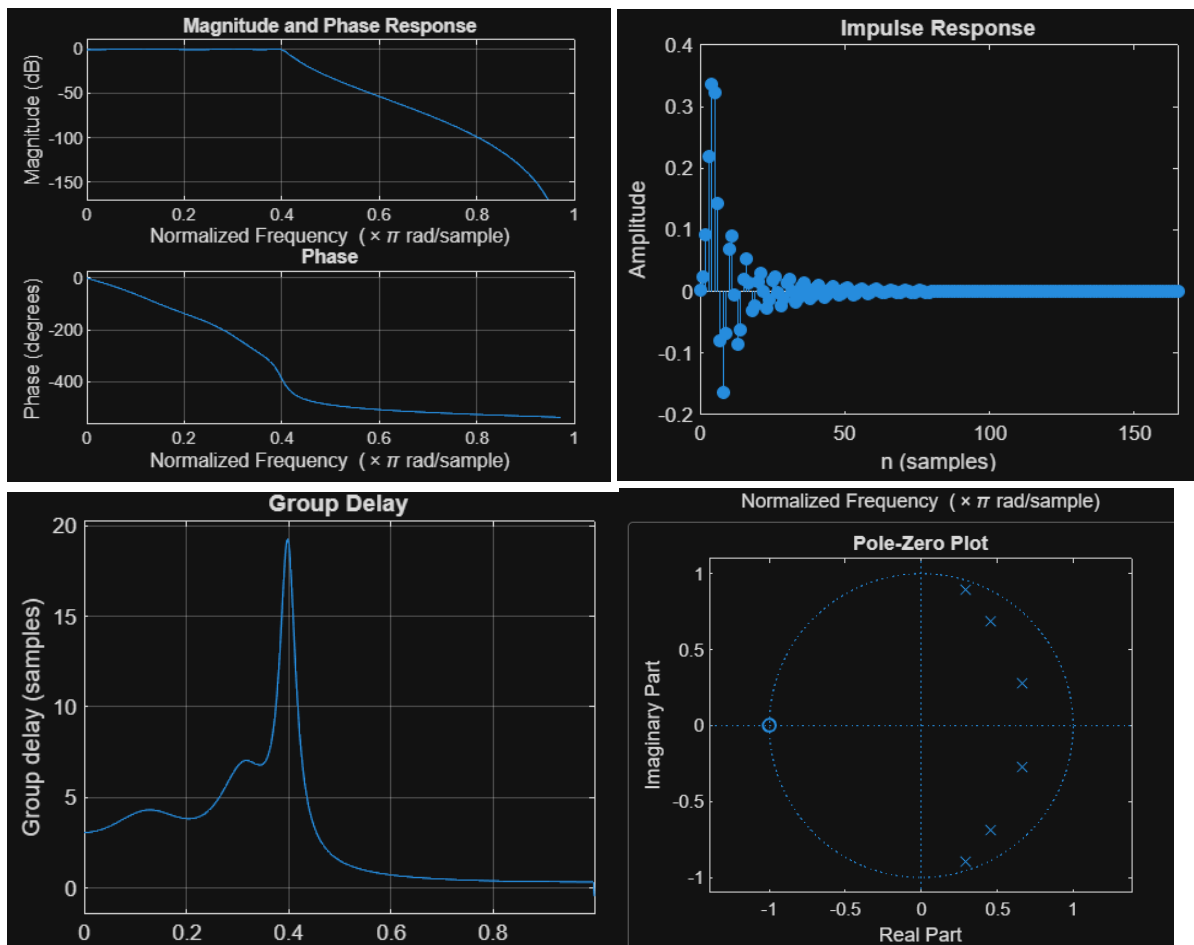
```
figure;
```

```
zplane(b,a);
```

```
grid on;
```

```
title('Pole-Zero Plot');
```

OUTPUT :-



Filter Order = 6

Cutoff Frequency = $0.40 * \pi$ rad/sample

System is STABLE

EX NO:2	Chebyshev IIR Filter
DATE:	2.2. Design and Performance Evaluation of a Chebyshev Type-I IIR High-Pass Filter

CODE :-

```

clc;

clear;

close all;

% Specifications

Rp = 1;

Rs = 40;

Wp = 0.5;

Ws = 0.35;

% Determine filter order

[N,Wn] = cheb1ord(Wp,Ws,Rp,Rs);

% Design High-Pass Filter

[b,a] = cheby1(N,Rp,Wn,'high');

% Display Results

fprintf('Filter Order = %d\n',N);

fprintf('Cutoff Frequency = %.4f\n',Wn);

if all(abs(roots(a))<1)

    disp('System is STABLE');

else

    disp('System is UNSTABLE');

end

%% Magnitude & Phase Response

figure;

freqz(b,a,1024);

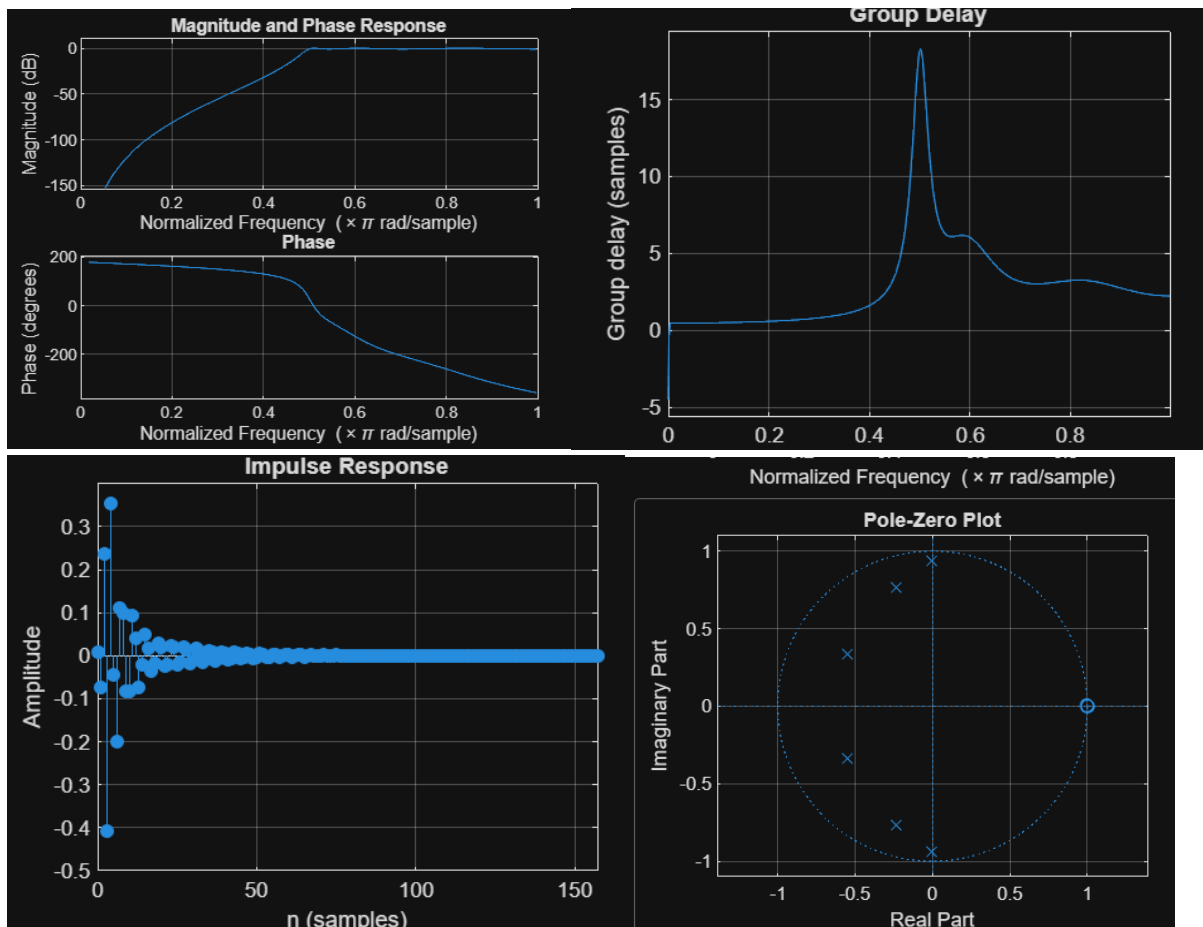
title('Magnitude and Phase Response');
```

```
%% Impulse Response
figure;
impz(b,a);
grid on;
title('Impulse Response');

%% Group Delay
figure;
grpdelay(b,a);
grid on;
title('Group Delay');

%% Pole-Zero Plot
figure;
zplane(b,a);
grid on;
title('Pole-Zero Plot');
```

OUTPUT :-



Filter Order = 6
Cutoff Frequency = 0.5000
System is STABLE